Design Patterns - Lab 2
Strategy

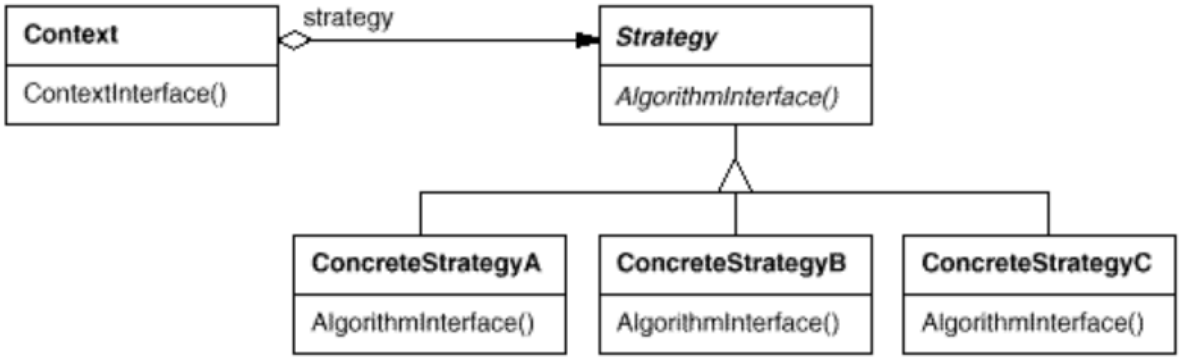
Goal
Introduce the basic concept of the Strategy pattern and apply the Strategy pattern to our book in order to format the alignment of the paragraphs.

Tasks
1. The users of our book writing software are willing to align the text of the paragraphs. Can we help them in this matter?

2. What would be the best way to implement an alignment for the text in the paragraph? Do you know any design pattern that can help?
   ……

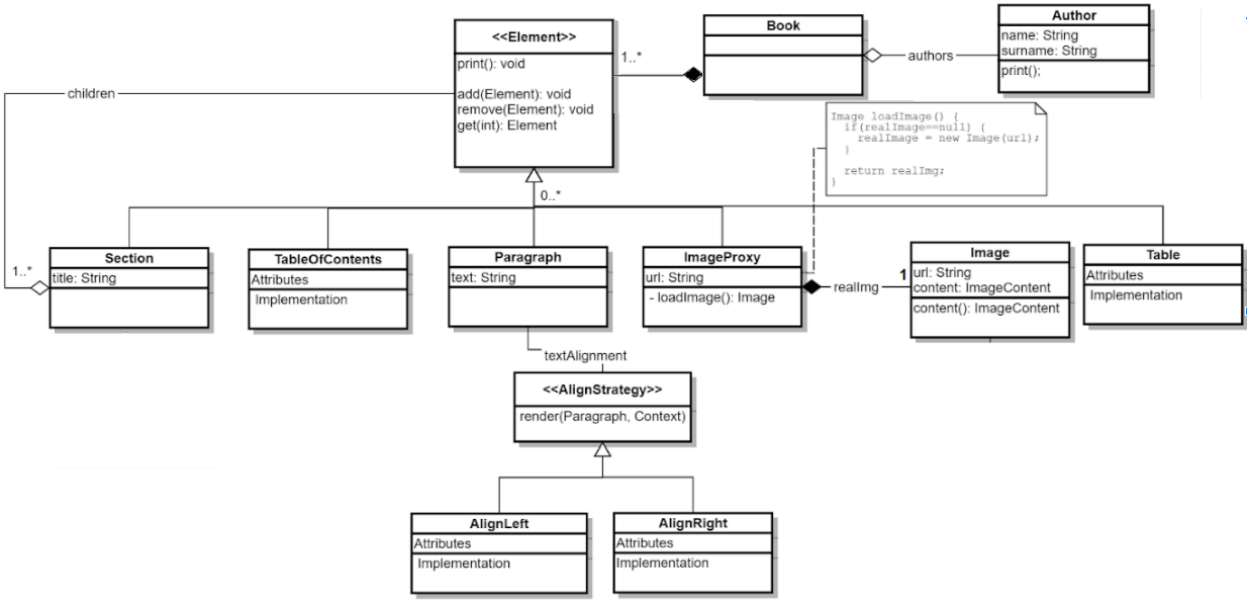
3. What about Strategy? Can we think of each of the alignment styles as a separate algorithm and implement it using the Strategy design pattern?
   ……

4. The diagram of the Strategy design pattern is:



5. How would our class view look if we add an AlignStrategy to our implementation?
   ……
   ……
   ……
   ……

6. Considering the following class diagram, implement the missing classes (ImageProxy is optional, not covered in this lab)



7. Add the main function to your implementation and add the missing implementation in order to compile and execute the main function code

```
public static void main(String[] args) throws Exception {
      Section cap1 = new Section("Capitolul 1");
      Paragraph p1 = new Paragraph("Paragraph 1");
      cap1.add(p1);
      Paragraph p2 = new Paragraph("Paragraph 2");
      cap1.add(p2);
      Paragraph p3 = new Paragraph("Paragraph 3");
      cap1.add(p3);
      Paragraph p4 = new Paragraph("Paragraph 4");
      cap1.add(p4);

      System.out.println("Printing without Alignment");
      System.out.println();
      cap1.print();

      p1.setAlignStrategy(new AlignCenter());
      p2.setAlignStrategy(new AlignRight());
      p3.setAlignStrategy(new AlignLeft());
```

```
        System.out.println();
        System.out.println("Printing with Alignment");
        System.out.println();
        cap1.print();
}
```

8. Discuss the design. What is the Context, the rendering context in render(Paragraph, Context) AlignStrategy's interface? What state does it holds?

9. List the advantages of the new design:
    a. Easy to add new paragraph formatters.
    b. ……
    c. ……

    List the liabilities of the new design:
    d. Coupling between AlignStrategy and Paragraph objects; can be loosen by changing render to `render(String, Context)`.

    > *Remember: Design (of any kind) is an iterative process and critical feedback is always a powerful tool.*

10. Don't forget to commit and push your code to your Github account.